

# STT 861: Theory of Probability & Statistics I

## Lecture 15: Discrete Distributions

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Reference: Casella & Berger Ch. 3.1–3.2

# Where we are: from tools to a catalogue of models

## The story so far

- **Ch. 1:** probability spaces, random variables, cdf, pmf/pdf
- **Ch. 2:** tools — transformations,  $\mathbb{E}g(X)$ ,  $\text{Var } X$ , mgf

So far every distribution was *given* to us. In practice, someone has to **choose** one.

## Chapter 3 roadmap

- 3.1 Introduction (today)
- 3.2 Discrete families (today)
- 3.3 Continuous families
- 3.4 Exponential families
- 3.5 Location–scale families
- 3.6 Inequalities & identities

## Goal for today

Understand what a **parametric family** is, why one number (or a few) can describe a whole random experiment, and how Chapter 2's tools plug into a family.

# Motivation: same question, many situations

| Situation                                | Random variable $X$        | Natural model             | Unknown         |
|------------------------------------------|----------------------------|---------------------------|-----------------|
| Does one patient respond to a treatment? | $X \in \{0, 1\}$           | Bernoulli( $p$ )          | $p$             |
| Defective chips among 20 inspected       | $X \in \{0, \dots, 20\}$   | Binomial( $20, \theta$ )  | $\theta$        |
| Calls to a help desk in one hour         | $X \in \{0, 1, 2, \dots\}$ | Poisson( $\lambda$ )      | $\lambda$       |
| Minutes until the next bus arrives       | $X > 0$                    | Exponential( $\beta$ )    | $\beta$         |
| Height of a randomly chosen adult        | $X \in \mathbb{R}$         | Normal( $\mu, \sigma^2$ ) | $\mu, \sigma^2$ |

## Key observation

The *shape* of the model is fixed by the story; only a few numbers (**parameters**) are unknown. Learning about the world = learning those numbers.

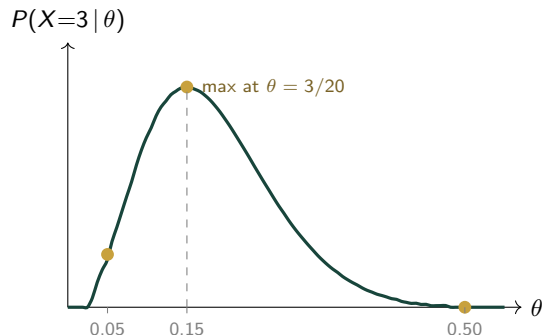
# Motivating example: why a family helps

A line inspects 20 chips and finds **3** defective. Each chip is defective independently with unknown probability  $\theta$ , so  $X \sim \text{Bin}(20, \theta)$ :

$$P(X = 3 | \theta) = \binom{20}{3} \theta^3 (1 - \theta)^{17} = 1140 \theta^3 (1 - \theta)^{17}.$$

| $\theta$ | $P(X = 3   \theta)$ |
|----------|---------------------|
| 0.05     | 0.060               |
| 0.15     | 0.243               |
| 0.50     | 0.001               |

Without a family we would need a pmf on  $\{0, \dots, 20\}$  (20 free numbers). With it, one number:  $\theta$ . The data point to  $\theta \approx 3/20$ .



The curve is the family evaluated at the observed data (preview of Ch. 7).

# Definition: a parametric family

## Parametric family of distributions

A family of pdfs or pmfs indexed by a parameter  $\theta$  is a collection

$$\{ f(x|\theta) : \theta \in \Theta \},$$

where  $\Theta \subseteq \mathbb{R}^k$  is the parameter space. Each fixed  $\theta$  gives one legitimate pdf/pmf in  $x$ ; changing  $\theta$  moves us to another member of the family.

## Examples of $\Theta$

- Bernoulli( $p$ ):  $\Theta = [0, 1]$
- Poisson( $\lambda$ ):  $\Theta = (0, \infty)$
- $N(\mu, \sigma^2)$ :  $\Theta = \mathbb{R} \times (0, \infty)$
- Unif( $a, b$ ):  $\Theta = \{(a, b) : a < b\}$

## Identifiability

We want  $\theta \neq \theta' \Rightarrow f(\cdot|\theta) \neq f(\cdot|\theta')$ .

**Counter-example:**  $X \sim N(\mu_1 + \mu_2, 1)$  with  $\theta = (\mu_1, \mu_2)$  cannot tell  $(1, 2)$  from  $(0, 3)$ . Fix: use  $\mu = \mu_1 + \mu_2$ .

# From the textbook: CB §3.1 and a notation convention

## CB §3.1 Introduction (summary; no theorems or examples in this section)

- Distributions are used to model **populations**; we usually deal with a **family** rather than a single distribution.
- The family is indexed by one or more **parameters**, which let us vary certain characteristics “while staying with one functional form”.
- Book’s example: the normal distribution may be a reasonable model, but we cannot precisely specify the mean. Then we deal with the **parametric family** of normal distributions with mean  $\mu$ , where  $\mu$  is unspecified,  $-\infty < \mu < \infty$ .
- Chapter 3 catalogues common distributions: mean, variance, typical applications and interrelationships (summarized in tables at the end of the book).

## Notation convention (CB §3.2)

Parameters are written in the pmf/pdf after a “|” (given), e.g.  $P(X = x | N)$  or  $f(x | \mu)$ . The same convention is used with cdfs, pdfs and expectations, and parameters may be dropped when there is no possibility of confusion.

## Example: the book's family — normal with unspecified mean

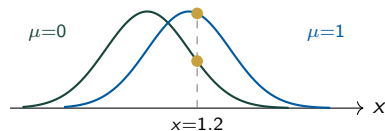
Model a population by  $N(\mu, 1)$  with  $\mu \in \Theta = \mathbb{R}$  unspecified:

$$f(x | \mu) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}.$$

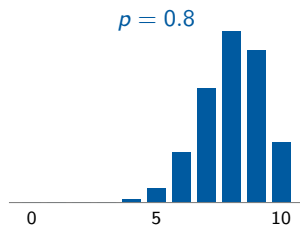
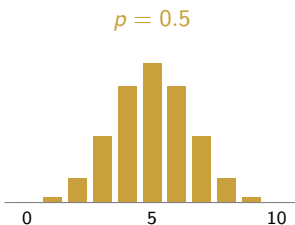
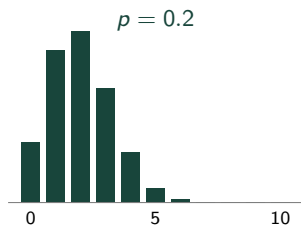
By Chapter 2:  $\mathbb{E}X = \mu$ ,  $\text{Var } X = 1$ ,  $M_X(t) = e^{\mu t + t^2/2}$ .  
One observation  $x = 1.2$  arrives. How plausible are two members of the family?

| $\mu$ | $f(1.2   \mu)$ |
|-------|----------------|
| 0     | 0.194          |
| 1     | 0.391          |

The member with  $\mu = 1$  explains the data about twice as well.



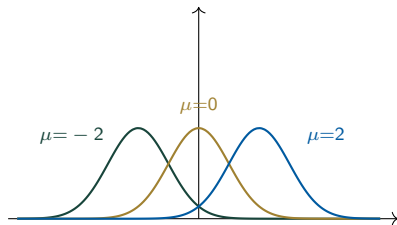
## Illustration: one family, three members — Binomial(10, $p$ )



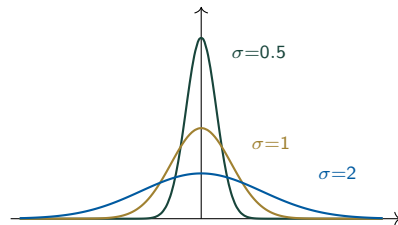
### Reading the picture

Same  $n = 10$ , same support  $\{0, \dots, 10\}$ , same formula  $\binom{10}{x} p^x (1-p)^{10-x}$ . Only  $p$  changes, yet the **location** and the **skewness** of the pmf change: right-skewed for small  $p$ , symmetric at  $p = \frac{1}{2}$ , left-skewed for large  $p$ .

## Illustration: what do parameters do? — Normal( $\mu, \sigma^2$ )



Changing  $\mu$  (location),  $\sigma = 1$



Changing  $\sigma$  (scale),  $\mu = 0$

### Preview of §3.5

$\mu$  slides the curve;  $\sigma$  stretches it (area always 1). Families built this way are *location-scale families*.

## Chapter 2 tools meet families: the Bernoulli( $p$ ) example

**Bernoulli( $p$ ):**  $P(X=1) = p$ ,  $P(X=0) = 1 - p$ ,  $p \in [0, 1]$

$$\mathbb{E}X = p, \quad \text{Var } X = \mathbb{E}X^2 - (\mathbb{E}X)^2 = p - p^2 = p(1 - p), \quad M_X(t) = \mathbb{E}e^{tX} = 1 - p + pe^t.$$

**Sum of iid Bernoullis  $\Rightarrow$  Binomial**

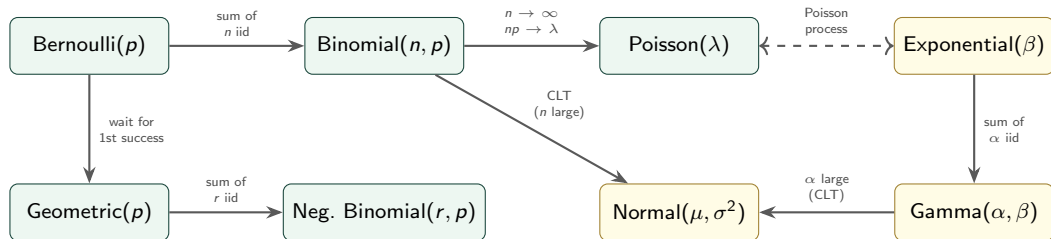
Let  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$  and  $S = \sum_{i=1}^n X_i$ . By independence,

$$M_S(t) = \prod_{i=1}^n M_{X_i}(t) = (1 - p + pe^t)^n,$$

which is the mgf of  $\text{Bin}(n, p)$ . By mgf uniqueness (CB Thm 2.3.11),  $S \sim \text{Bin}(n, p)$ . Hence  $\mathbb{E}S = np$  and  $\text{Var } S = np(1 - p)$ .

**Moral:** moments and mgf of a family are **functions of the parameter**; compute them once, use them forever.

# Families are related: a map of Chapter 3



**Green:** discrete (§3.2). **Gold:** continuous (§3.3). Solid arrows: derivations we will prove this term. Dashed: two views of one random phenomenon.

# Why study families, and our checklist for each one

## Three reasons

1. **Modelling:** a physical story picks the family.
2. **Computation:** moments, mgf and relations are derived once.
3. **Inference:** unknown  $\theta$  is what we estimate from data (Stat II).

## For every family we learn. . .

- the pmf/pdf and the support
- parameters and the space  $\Theta$
- $\mathbb{E}X$ ,  $\text{Var } X$ ,  $M_X(t)$
- the story behind it
- how it connects to other families

## What is coming next

**§3.2 Discrete (second half of today):** discrete uniform, hypergeometric, binomial, Poisson, negative binomial, geometric.

**§3.3 Continuous (next lecture):** uniform, gamma (exponential,  $\chi^2$ ), normal, beta, Cauchy, lognormal, double exponential.

# Activity 1 — Match the story to a family (5 min, pairs)

## Activity 1

Choose from: Bernoulli, Binomial, Poisson, Geometric, Exponential, Normal, Discrete uniform. Name the family *and* the unknown parameter(s).

- (a) Number of customers entering a shop in 10 minutes.
- (b) Number of 10 patients who respond to a treatment, each independently with probability  $p$ .
- (c) Minutes until the next bus arrives.
- (d) Number of coin tosses until the first head.
- (e) Heights of adult women in a population.
- (f) The face showing on one roll of a fair six-sided die.

**Discuss:** in (f), is there any unknown parameter left?

# Activity 1 — Solutions

- Ⓐ Poisson( $\lambda$ );  $\lambda > 0$  is the mean number of arrivals per 10 minutes.
- Ⓑ Binomial( $10, p$ );  $p \in [0, 1]$  (here  $n = 10$  is known).
- Ⓒ Exponential( $\beta$ );  $\beta > 0$  is the mean waiting time (CB parametrization).
- Ⓓ Geometric( $p$ ) on  $\{1, 2, \dots\}$ ;  $p \in (0, 1]$ , the probability of a head.
- Ⓔ Normal( $\mu, \sigma^2$ );  $\mu \in \mathbb{R}$ ,  $\sigma^2 > 0$ .
- Ⓕ Discrete uniform on  $\{1, \dots, 6\}$ ,  $P(X = k) = 1/6$ . No unknown parameter:  $N = 6$  is known, so this is a single distribution, not a family with free parameters.

## Activity 2 — Parameter spaces and identifiability (8 min, groups of 3)

### Activity 2

- Ⓐ Write the parameter space  $\Theta$  for each: Bernoulli( $p$ ), Binomial( $n, p$ ), Poisson( $\lambda$ ), Exponential( $\beta$ ), Unif( $a, b$ ).
- Ⓑ Is  $\{f(x|\theta) = \theta x^{\theta-1}, 0 < x < 1 : \theta > 0\}$  a legitimate family? Verify that each member integrates to 1 and compute  $\mathbb{E}X$ .
- Ⓒ Which of these is a *parametric* family? (i)  $\{\text{Bin}(10, p) : p \in [0, 1]\}$  (ii) all pdfs on  $(0, 1)$  (iii)  $\{N(\mu, 1) : \mu \in \mathbb{R}\}$
- Ⓓ  $X \sim N(\mu_1\mu_2, 1)$  with  $\theta = (\mu_1, \mu_2) \in (0, \infty)^2$ . Is  $\theta$  identifiable? If not, reparametrize.

## Activity 2 — Solutions

- Ⓐ Bernoulli:  $[0, 1]$ . Binomial:  $\{(n, p) : n \in \{1, 2, \dots\}, p \in [0, 1]\}$ . Poisson:  $(0, \infty)$ . Exponential:  $(0, \infty)$ . Unif( $a, b$ ):  $\{(a, b) \in \mathbb{R}^2 : a < b\}$ .
- Ⓑ For every  $\theta > 0$ :  $f \geq 0$  and  $\int_0^1 \theta x^{\theta-1} dx = [x^\theta]_0^1 = 1$ . So yes: this is Beta( $\theta, 1$ ).  $\mathbb{E}X = \int_0^1 \theta x^\theta dx = \frac{\theta}{\theta + 1}$ .
- Ⓒ (i) and (iii) are parametric (finitely many real parameters). (ii) is *nonparametric*: it is not indexed by a finite-dimensional  $\theta$ .
- Ⓓ Not identifiable:  $(\mu_1, \mu_2) = (1, 6)$  and  $(2, 3)$  give the same  $N(6, 1)$ . Only the product  $\mu = \mu_1 \mu_2 \in (0, \infty)$  is identifiable; use  $\theta = \mu$ .

## Activity 3 — Bernoulli and Binomial via the mgf (10 min)

### Activity 3

Let  $X \sim \text{Bernoulli}(p)$ .

- Ⓐ Find  $\mathbb{E}X$ ,  $\text{Var } X$ , and  $M_X(t)$ .
- Ⓑ For which  $p$  is  $\text{Var } X$  largest, and what is that maximum? Interpret.
- Ⓒ Let  $X_1, \dots, X_4 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$  and  $S = \sum X_i$ . Find  $M_S(t)$  and name the distribution of  $S$ .
- Ⓓ If  $p = \frac{1}{2}$ , compute  $P(S = 2)$  directly from the Binomial pmf.

## Activity 3 — Solutions

- Ⓐ  $\mathbb{E}X = p$ ,  $\text{Var } X = p(1 - p)$ ,  $M_X(t) = 1 - p + pe^t$ .
- Ⓑ  $\frac{d}{dp} p(1 - p) = 1 - 2p = 0 \Rightarrow p = \frac{1}{2}$  (concave, so a maximum), with  $\text{Var } X = \frac{1}{4}$ . A fair coin is the most “unpredictable”; as  $p \rightarrow 0$  or 1 the outcome becomes nearly certain.
- Ⓒ  $M_S(t) = (1 - p + pe^t)^4$ , so  $S \sim \text{Bin}(4, p)$  by mgf uniqueness.
- Ⓓ  $P(S = 2) = \binom{4}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{6}{16} = \frac{3}{8}$ .